

Vaibhav Chaubey

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EDUCATION

B.Tech in Computer Science & Engineering

IIIT Raichur, India

Expected May 2027

GPA: 7.3/10.0

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

Languages

JavaScript (ES2022), TypeScript, Python, C, C++, SQL, GraphQL

AI & LLM

Anthropic Claude API, OpenAI API, RAG pipelines, Qdrant, LangGraph, MCP, SSE Streaming, Token optimization, Prompt engineering, Ollama

Frontend

React.js (18/19), React Router 7, Redux, Vite (IIFE SDK), Tailwind CSS 4, Axios

Backend

Node.js, Express.js, FastAPI, Prisma ORM, JWT, Webhooks, Producer-consumer pipelines, Socket.IO, Cron jobs

Databases & Infra

MySQL, MongoDB, Qdrant, Docker, Bitbucket Pipelines, Linux, Shopify Admin/Billing API, Cloudinary

Architecture

Multi-tenant SaaS, Dual-layer caching, Provider-agnostic LLM abstraction, Rate limiting, State machines, IIFE widget SDKs, WebSocket gateways

ACHIEVEMENTS & INTERESTS

Problem Solving: Solved 200+ DSA challenges across LeetCode, HackerRank, and GeeksforGeeks

Interests: Agentic AI Architecture, LLM Systems Design, Competitive Programming

EXPERIENCE

Full-Stack Developer Intern (MERN + AI/LLM) | Setubridge Technolabs, Ahmedabad

Dec 2025 – Present

AI Shopping Assistant — Production Multi-Tenant Shopify SaaS (Sole Engineer)

React Router 7 · Node.js · TypeScript · Prisma/MySQL · Qdrant · Socket.IO · Anthropic Claude · OpenAI · Docker · DigitalOcean

- Shipped a 3-service production Shopify app — Remix backend, a JWT-authenticated WebSocket gateway (Socket.IO), and a separate React agent dashboard — serving merchants on DigitalOcean with per-shop row-level isolation across 26 Prisma models and 66 REST routes.
- Cut LLM cost per conversation ~55% by replacing a monolithic prompt with a 7-state intent engine that loads tier-based prompt modules on demand; classification uses cosine-scored embedding prototypes across 3 confidence bands with an LRU-cached GPT-4o-mini fallback (500 entries, 1500 ms timeout) bounding cost on repeat ambiguous queries.
- Built end-to-end bot-to-human handoff over WebSockets — JWT-authenticated agents, atomic SQL-level conversation claim preventing double-claim races, shared inbox across all agents in a shop, multi-tab presence dedupe, and reconnect catchup of messages missed during 2-min disconnect windows.
- Designed a provider-agnostic LLM gateway over Anthropic Claude and an OpenAI-compatible bus (OpenAI, Groq, Together, Gemini) with unified SSE streaming, tool-schema normalization, prompt-cache hit-rate monitoring tied to microdollar-precision cost accounting, and recovery from malformed failed_generation tool calls.
- Integrated Shopify Storefront MCP behind a two-tier cache (10-min tool-discovery + 5-min result TTL) saving ~250 ms/message, and a Qdrant RAG pipeline over 5 source types (page/url/pdf/text/faq) with batched multi-provider embeddings.

OpsPanel — Internal Shopify Pricing & Subscription Platform

React 19 · Express 5 · MySQL · Vite · Shopify Billing API · JWT · Bitbucket Pipelines

- Built an internal full-stack platform to manage pricing plans, subscriptions, and billing for company Shopify apps via Shopify Billing API (GraphQL); multi-tenant architecture with dual-auth (JWT + stateless Shopify token validation), per-shop rate limiting (100 req/min), and a 5-min token cache reducing external API calls by ~80%.
- Developed embeddable pricing widgets (React + Vite IIFE bundles) with MutationObserver auto-mount and cross-widget communication, enabling 10 composable components to deploy on any Shopify storefront via a single <div> tag; integrated Shopify Billing API with a 6-state subscription machine and automated trial-expiry cron.
- Built an admin dashboard covering campaign management, coupon engine, CMS, trial tracking, and automated merchant provisioning — backed by 16 MySQL tables across 15 REST modules (~80K LOC).

AI Demo-Video Generator — Narrated Screen Walkthrough → Auto-Voiced Marketing Video (Built End-to-End)

Chrome MV3 (vanilla JS) · Node/TypeScript/Express · OpenAI GPT-4o (structured outputs) + Whisper · Python · FFmpeg · edge-tts · Caddy + systemd

- Engineered a Chrome MV3 recorder that makes understanding the screen a JSON merge instead of computer vision — synchronizing tab video (VP9 8 Mbps), mic narration, a live Web Speech transcript, and capture-phase DOM events (each carrying a robust data-test/aria/#id/nth-of-type selector, visible label, and URL) across a stateless service worker, a durable offscreen document, and a content script onto ONE absolute clock (performance.timeOrigin + performance.now()) via an explicit readiness handshake, not fixed delays.
- Built a token-gated Node/TypeScript Express pipeline that fuses events + transcript into a heuristic 8–12-scene timeline (scroll-coalescing, idle-gap segmentation), then drives GPT-4o structured outputs (strict json_schema) on a hallucination-guarded prompt grounded only in real on-screen labels — a timestamp/scene-parity validator auto re-prompts once before a deterministic fallback, and empty in-browser transcripts are backfilled by server-side Whisper aligned to the recording clock.
- Architected a Python/FFmpeg compositor with scene-level, action-anchored time-warp — each scene's video is warped (setpts, clamped 0.5x–3x) toward its natural-rate edge-tts voiceover while residual mismatch is absorbed as trailing silence or a held-frame freeze, eliminating chipmunk audio, cut-off sentences, and cumulative drift by construction; deployed on a Linux VM (Caddy HTTPS, systemd, sequential FFmpeg queue) so a non-technical teammate self-serves a finished MP4 in ~30–60s at cents/video.

TECHNICAL PROJECTS

HireScope — AI Interview Assistant | React, Redux, Node.js/Express, Gemini API

May – Oct 2025

- Built a two-tab AI interview app — interviewee chat tab and a live-synced interviewer dashboard — with PDF/DOCX resume parsing (Name/Email/Phone extraction) and auto-prompting for missing fields; state persisted across reloads via redux-persist.
- Gemini-powered interview engine generates 6 role-specific questions with configurable auto-submit timers (20s/60s/120s); scores answers, produces summaries, and surfaces candidates ranked by score with per-candidate chat history and search/sort in the recruiter dashboard.

Placement Website — Placement Management Portal | React, Node.js, MongoDB,

2025

Cloudinary, JWT

- Full-stack placement management platform with JWT auth, Cloudinary resume upload/download, and bulk admin workflows — CSV student import and ZIP export of resumes.
- Company CRUD, application tracking with advanced filters (on/off-campus, remote/hybrid), and a coordinator dashboard with key placement statistics.